RH_CHECKPOINT – Full Technical Master v3.2

Transport–Spectral Contradiction Program for the Riemann Hypothesis

Closure, Drift, No-Hiding, Packet Coercivity, Gaussian Regularization, and Model Bricks

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April 2026

Abstract

We present a sharpened localized transport–spectral checkpoint for the Riemann Hypothesis. The framework isolates a precise conditional contradiction inside a dipole-regularized residue channel localized around a single candidate off-critical zero

$$\rho = \beta + i\gamma, \qquad \beta \neq \frac{1}{2}.$$

On the closure side, a localized exact window law forces the localized excess energy to vanish:

$$E_{\mathrm{ex}}^{(\rho)}(f) = 0.$$

On the obstruction side, a transported drift law

$$\delta_\rho(u) = a_\rho u + O(1), \qquad a_\rho = \frac{\beta - \frac{1}{2}}{2\pi} \neq 0,$$

generates a quartic penalty

$$P_4(\rho; \tau, L) \gtrsim \tau^4 L.$$

If the localized witness bank detects this drift coercively and cross/remainder leakage remains subcritical, then the same localized channel satisfies

$$E_{\mathrm{ex}}^{(\rho)}(f) > 0,$$

contradicting closure.

This v3.2 master version keeps the main proof chain short but preserves all essential technical details: Fourier preliminaries, dipole regularization, localized witness banks, exact-window closure, quartic obstruction, bridge inequality, bank-subcritical remainder, finite Gaussian Gram positivity, windowed packet coercivity, perturbative stability, nonstationary cross-term decay, Gaussian heat-semigroup regularization, and the Source–Love Channel as a separate meta-model.

The document remains strictly conditional. It is not an unconditional proof of RH. It reduces RH to explicit analytic bottlenecks on the genuine zeta residue channel and supplies rigorous model bricks showing that the required mechanisms are mathematically nonempty.

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1 Public Clarification

Important clarification. This document is a conditional checkpoint, not an unconditional proof of the Riemann Hypothesis. It isolates a precise contradiction mechanism and reduces RH to explicit analytic tasks that must still be verified on the genuine zeta residue channel. All model results in the appendices are proved only in controlled regimes; their transfer to the actual explicit-formula channel remains open.

Důležité upřesnění. Tento dokument není nepodmíněný důkaz Riemannovy hypotézy. Jde o podmíněný checkpoint, který izoluje přesný mechanismus rozporu a převádí RH na explicitní analytické úkoly na skutečném zeta-reziduálním kanálu. Všechny modelové výsledky v dodatcích jsou dokázány pouze v kontrolovaných modelech; jejich přenos na skutečný kanál explicitní formule zůstává otevřený.

Remark 1.1. *This clarification is part of the logical status of the paper. The model appendices are not promoted to genuine zeta-side theorems. The RH implication requires verifying the corresponding estimates for the actual residue channel.*

2 Introduction

The Riemann Hypothesis asserts that every nontrivial zero

$$\rho = \beta + i\gamma$$

of the Riemann zeta function satisfies

$$\beta = \frac{1}{2}.$$

The present framework does not attempt to locate all zeros simultaneously. Instead, it asks a localized question:

Would the existence of even a single off-critical zero necessarily create a detectable energetic obstruction in a suitably localized readout channel?

The intended contradiction has two sides:

$$\boxed{\text{localized exact window law} \Longrightarrow E_{\mathrm{ex}}^{(\rho)}(f) = 0}$$

and

$$\boxed{\text{transported drift + witness coercivity + no-hiding} \Longrightarrow E_{\mathrm{ex}}^{(\rho)}(f) > 0.}$$

Thus the program reduces RH to the verification of a localized exact window law and a localized bridge package for every candidate off-critical zero-channel.

The document is organized as follows. The main body gives definitions, assumptions, and the fixed-channel contradiction. The appendices contain the model bricks: soft readout survival, pure exponential coercivity, windowed packet coercivity, cross-term decay, Gaussian regularization, and the Source–Love Channel meta-model.

3 Why Localization Matters

Localization provides three advantages over a purely global approach.

(i) **Single-channel obstruction.** A candidate zero ρ is treated as an individual transported mode, rather than as an anonymous contribution to the global zero set.

(ii) **Explicit interference control.** The interaction between the candidate mode and the remainder becomes a localized cross term that can be attacked by nonstationary-phase estimates.

(iii) **Modular bottlenecks.** The proof decomposes into independent analytic tasks: transported drift, witness coercivity, no-hiding leakage, bank-subcritical remainder, Gram stability, exact window closure, and model-to-zeta transfer.

Localization does not weaken the statement. It sharpens the target.

4 Public-Facing Synthesis, Made Proof-Safe

4.1 Compressed description

If an off-critical zero exists, then after suitable transport it should generate a localized linear drift. A witness bank should detect this drift as a coercive quartic obstruction. If remainder and cross terms are subcritical, the localized excess becomes positive. A localized exact window law would force the same excess to vanish. The contradiction excludes the off-critical channel once all assumptions are verified on the genuine zeta residue side.

In one line:

$$\text{localize the channel} \longrightarrow \text{read the drift} \longrightarrow \text{keep leakage subcritical} \longrightarrow \text{close the window.}$$

4.2 Claim hygiene

H1. Do not present the framework as an unconditional proof of RH.

H2. Do not promote model lemmas to statements about the genuine zeta residue channel.

H3. Do not treat visibility, screenshots, or social reception as mathematical evidence.

H4. Keep the proof chain fixed:

$$\text{drift} + \text{coercivity} + \text{no-hiding leakage} + \text{exact window} \Longrightarrow \text{contradiction.}$$

H5. Separate exposition and proof components. Symbolic language may guide intuition but cannot replace estimates.

5 Dependency Ledger

The document separates four logical layers.

Layer	Content	Status
Definitions	localized bank, excess, window, drift, penalty	structural
Algebraic implications	closure, bridge inequality, fixed-channel contradiction	proved here
Model bricks	soft survival, packet coercivity, cross-term decay, Gaussian hammer	proved in model
Zeta-side tasks	drift law, coercivity, leakage, Gram bounds, exact window	open

The proof chain is not

$$\text{model lemma} \Longrightarrow \text{RH.}$$

It is

$$\text{model lemma} \Longrightarrow \text{evidence that the analytic target is nonempty.}$$

The actual RH implication requires corresponding statements on the genuine zeta residue channel.

6 Assumption Ledger for the Fixed Channel

Fix a candidate zero

$$\rho = \beta + i\gamma, \qquad \beta \neq \frac{1}{2}.$$

The fixed-channel contradiction uses the following assumptions.

A1. Transported drift. There exists a transported phase/readout variable

$$\delta_\rho(u) = a_\rho u + r_\rho(u), \qquad a_\rho = \frac{\beta - \frac{1}{2}}{2\pi} \neq 0,$$

where r_ρ is bounded.

A2. Localized exact window law. The localized witness-bank energy satisfies

$$Q^{(\rho)}_{\text{bank}}(f;\tau,L) = E^{(\rho)}_0 L + J^{(\rho)}_f(\tau+L) - J^{(\rho)}_f(\tau) + R^{(\rho)}(\tau,L),$$

with $J^{(\rho)}_f$ bounded and

$$\lim_{L\to\infty} \sup_{\tau\in\mathbb{R}} \frac{|R^{(\rho)}(\tau,L)|}{L} = 0.$$

A3. Localized witness coercivity. There exist $\kappa > 0$ and $C_0 \geq 0$ such that

$$Q^{(\rho)}_\rho(\tau,L) \geq \kappa P_4(\rho;\tau,L) - C_0 L.$$

A4. Localized no-hiding leakage. There exist $C_1 \geq 0$ and $\eta(\tau) \to 0$ such that

$$Q^{(\rho)}_{\text{cross}}(\tau,L) + Q^{(\rho)}_{\text{rest}}(\tau,L) \geq -\eta(\tau) P_4(\rho;\tau,L) - C_1 L.$$

Only A1–A4 enter the fixed-channel contradiction. The model-to-zeta transfer remains open.

7 Fourier Preliminaries

We use the Fourier transform convention

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x\xi}\, \mathrm{d}x.$$

Let

$$\psi(x) = e^{-\pi x^2}.$$

Lemma 7.1 (Gaussian self-duality)**.** *With the above convention,*

$$\widehat{\psi} = \psi.$$

Proof. Let

$$F(\xi) = \widehat{\psi}(\xi).$$

Differentiating under the integral sign,

$$F'(\xi) = \int_{\mathbb{R}} e^{-\pi x^2}(-2\pi i x) e^{-2\pi i x\xi}\, \mathrm{d}x.$$

Since

$$\frac{\mathrm{d}}{\mathrm{d}x} e^{-\pi x^2} = -2\pi x e^{-\pi x^2},$$

we get

$$F'(\xi) = i \int_{\mathbb{R}} \frac{\mathrm{d}}{\mathrm{d}x}(e^{-\pi x^2}) e^{-2\pi i x\xi}\, \mathrm{d}x.$$

Integration by parts gives

$$F'(\xi) = -2\pi\xi F(\xi).$$

Thus

$$F(\xi) = C e^{-\pi\xi^2}.$$

At $\xi = 0$,

$$C = F(0) = \int_{\mathbb{R}} e^{-\pi x^2}\, \mathrm{d}x = 1.$$

Therefore

$$F(\xi) = e^{-\pi \xi^2}.$$

□

Since the Fourier transform satisfies

$$\mathcal{F}^4 = I$$

on $L^2(\mathbb{R})$, its spectral values lie in

$$\{1, -i, -1, i\}.$$

For each

$$\lambda \in \{1, -i, -1, i\},$$

define

$$P_\lambda = \frac{1}{4} \sum_{m=0}^{3} \lambda^{-m} \mathcal{F}^m.$$

Proposition 7.2 (Fourier spectral projectors)**.** *For every $\lambda \in \{1, -i, -1, i\}$,*

$$P_\lambda^2 = P_\lambda, \qquad \mathcal{F}(P_\lambda f) = \lambda P_\lambda f.$$

In particular,

$$P_{+1} = \frac{1}{4}(I + \mathcal{F} + \mathcal{F}^2 + \mathcal{F}^3)$$

projects onto the Fourier-fixed self-dual sector.

Proof. The identity $\mathcal{F}^4 = I$ implies that the minimal polynomial of $\mathcal{F}$ divides

$$z^4 - 1.$$

The operators P_λ are the finite spectral projectors for the cyclic group generated by $\mathcal{F}$.

Directly,

$$\mathcal{F} P_\lambda = \frac{1}{4} \sum_{m=0}^{3} \lambda^{-m} \mathcal{F}^{m+1}.$$

Reindexing modulo 4, and using $\lambda^4 = 1$, gives

$$\mathcal{F} P_\lambda = \lambda P_\lambda.$$

For idempotence, use finite Fourier orthogonality:

$$\frac{1}{4} \sum_{m=0}^{3} (\lambda^{-1}\mu)^m = \begin{cases} 1, & \lambda = \mu, \\ 0, & \lambda \neq \mu. \end{cases}$$

Hence $P_\lambda P_\mu = 0$ for $\lambda \neq \mu$, and $P_\lambda^2 = P_\lambda$. □

Remark 7.3 (Fourier as phase accounting)**.** *In the Hermite basis $\{h_n\}_{n \geq 0}$,*

$$\mathcal{F} h_n = (-i)^n h_n.$$

Thus Fourier evolution is pure phase cycling on Hermite modes:

$$1,\ -i,\ -1,\ i.$$

In the correct basis, Fourier does not mix structure chaotically. It rotates phase mode by mode.

8 Dipole-Regularized Residue Channel

Let μ denote the residue distribution arising from a fixed explicit-formula normalization. Let

$$\psi \in \mathcal{S}(\mathbb{R})$$

be a Schwartz test function satisfying the dipole condition

$$\widehat{\psi}(0) = 0.$$

Define the dipole-regularized residue channel

$$f = \psi * \mu.$$

The dipole condition removes the constant component and leaves a fluctuation channel suitable for second-difference witness tests.

For $a > 0$, define the second-difference witness operator

$$(W_a h)(u) = h(u+a) - 2h(u) + h(u-a).$$

Remark 8.1. *The operator W_a annihilates affine functions and detects curvature at scale a. This is why it is suited to measuring drift-induced obstruction.*

9 Localized Witness Bank

Fix a candidate zero

$$\rho = \beta + i\gamma.$$

Let $w_\rho(u) \geq 0$ be a localization window. Let

$$a_j > 0, \qquad j = 0, \ldots, J,$$

be finitely many witness scales, and let $w_j > 0$ be positive bank weights.

Definition 9.1 (Localized witness bank)**.** *The ρ-localized witness-bank energy is*

$$Q^{(\rho)}_{\text{bank}}(f;\tau,L) = \sum_{j=0}^{J} w_j^2 \int_\tau^{\tau+L} w_\rho(u)|W_{a_j} f(u)|^2 \, \mathrm{d}u.$$

Definition 9.2 (Localized excess)**.** *Given a baseline density $E_0^{(\rho)} \geq 0$, define*

$$E^{(\rho)}_{\text{ex}}(f) = \limsup_{L\to\infty} \sup_{\tau\in\mathbb{R}} \frac{Q^{(\rho)}_{\text{bank}}(f;\tau,L) - E_0^{(\rho)} L}{L}.$$

Assume a single-zero splitting

$$f = f_\rho + f_{\text{rest}}.$$

Then

$$W_{a_j} f = W_{a_j} f_\rho + W_{a_j} f_{\text{rest}}.$$

Therefore

$$Q^{(\rho)}_{\text{bank}}(f;\tau,L) = Q^{(\rho)}_{\rho}(\tau,L) + Q^{(\rho)}_{\text{cross}}(\tau,L) + Q^{(\rho)}_{\text{rest}}(\tau,L),$$

where

$$Q^{(\rho)}_{\rho}(\tau,L) = \sum_{j=0}^{J} w_j^2 \int_\tau^{\tau+L} w_\rho(u)|W_{a_j} f_\rho(u)|^2 \, \mathrm{d}u,$$

$$Q^{(\rho)}_{\text{cross}}(\tau,L) = 2\sum_{j=0}^{J} w_j^2 \operatorname{Re} \int_\tau^{\tau+L} w_\rho(u) W_{a_j} f_\rho(u) \overline{W_{a_j} f_{\text{rest}}(u)} \, \mathrm{d}u,$$

and

$$Q^{(\rho)}_{\text{rest}}(\tau,L) = \sum_{j=0}^{J} w_j^2 \int_\tau^{\tau+L} w_\rho(u)|W_{a_j} f_{\text{rest}}(u)|^2 \, \mathrm{d}u.$$

10 Localized Exact Window Law

Definition 10.1 (Localized exact window law)**.** *We say that the ρ-localized exact window law holds if there exist a bounded function*

$$J_f^{(\rho)} : \mathbb{R} \to \mathbb{R}$$

and a remainder $R^{(\rho)}(\tau, L)$ such that

$$Q_{\text{bank}}^{(\rho)}(f; \tau, L) = E_0^{(\rho)} L + J_f^{(\rho)}(\tau + L) - J_f^{(\rho)}(\tau) + R^{(\rho)}(\tau, L),$$

where

$$\sup_{\tau \in \mathbb{R}} |J_f^{(\rho)}(\tau)| < \infty,$$

and

$$\lim_{L \to \infty} \sup_{\tau \in \mathbb{R}} \frac{|R^{(\rho)}(\tau, L)|}{L} = 0.$$

Theorem 10.2 (Localized closure)**.** *If the localized exact window law holds, then*

$$E_{\text{ex}}^{(\rho)}(f) = 0.$$

Proof. Divide the exact window identity by L:

$$\frac{Q_{\text{bank}}^{(\rho)}(f; \tau, L) - E_0^{(\rho)} L}{L} = \frac{J_f^{(\rho)}(\tau + L) - J_f^{(\rho)}(\tau)}{L} + \frac{R^{(\rho)}(\tau, L)}{L}.$$

Taking absolute values and then the supremum over τ,

$$\sup_{\tau} \left| \frac{Q_{\text{bank}}^{(\rho)}(f; \tau, L) - E_0^{(\rho)} L}{L} \right| \leq \frac{2\|J_f^{(\rho)}\|_\infty}{L} + \sup_{\tau} \frac{|R^{(\rho)}(\tau, L)|}{L}.$$

Both terms tend to 0 as $L \to \infty$. Hence

$$E_{\text{ex}}^{(\rho)}(f) = 0.$$

□

11 Transported Drift and Quartic Obstruction

Definition 11.1 (Transported drift law)**.** *For a candidate zero*

$$\rho = \beta + i\gamma,$$

we say that the transported drift law holds if

$$\delta_\rho(u) = a_\rho u + r_\rho(u), \qquad a_\rho = \frac{\beta - \frac{1}{2}}{2\pi},$$

where r_ρ is bounded.

Definition 11.2 (Quartic penalty)**.** *Define*

$$P_4(\rho; \tau, L) = \int_\tau^{\tau+L} |\delta_\rho(u)|^4 \, \mathrm{d}u.$$

Lemma 11.3 (Quartic lower bound)**.** *Assume the transported drift law and $a_\rho \neq 0$. Then for sufficiently large τ and all $L \geq 1$,*

$$P_4(\rho; \tau, L) \geq c_\rho \tau^4 L$$

for some constant $c_\rho > 0$.

Proof. Write

$$\delta_\rho(u) = a_\rho u + r_\rho(u), \qquad |r_\rho(u)| \le M.$$

For sufficiently large u,

$$|\delta_\rho(u)| \ge |a_\rho| u - M \ge \frac{|a_\rho|}{2} u.$$

Thus for $u \in [\tau, \tau + L]$ and τ sufficiently large,

$$|\delta_\rho(u)|^4 \ge \frac{|a_\rho|^4}{16} u^4 \ge \frac{|a_\rho|^4}{16} \tau^4.$$

Integrating over an interval of length L,

$$P_4(\rho; \tau, L) \ge \frac{|a_\rho|^4}{16} \tau^4 L.$$

□

Lemma 11.4 (Polynomial upper bound for quartic penalty)**.** *Assume the transported drift law and $|r_\rho(u)| \le M$. Then there exists $C_\rho > 0$ such that*

$$P_4(\rho; \tau, L) \le C_\rho \Big((\tau + L)^5 - \tau^5 + L \Big)$$

for all $\tau \ge 1$, $L \ge 1$.

Proof. Since

$$|\delta_\rho(u)| \le |a_\rho| u + M,$$

there is a constant $C > 0$ such that

$$|\delta_\rho(u)|^4 \le C(u^4 + 1).$$

Therefore

$$P_4(\rho; \tau, L) \le C \int_\tau^{\tau+L} (u^4 + 1) \, \mathrm{d}u = C \left(\frac{(\tau + L)^5 - \tau^5}{5} + L \right).$$

Absorb the factor $1/5$ into the constant. □

Remark 11.5 (Why $m_0 < 2$ is critical)**.** *The quartic obstruction grows like*

$$P_4(\rho; \tau, L) \gtrsim \tau^4 L.$$

If a remainder contribution grows like

$$Q_{\mathrm{rest}}^{(\rho)}(\tau, L) \lesssim \tau^{2m_0} L,$$

then

$$\frac{Q_{\mathrm{rest}}^{(\rho)}}{P_4} \lesssim \tau^{2m_0 - 4}.$$

Thus $m_0 < 2$ is exactly the condition $2m_0 < 4$, which forces the ratio to vanish.

12 Localized Bridge Package

Definition 12.1 (Localized witness coercivity)**.** *The localized bank is coercive on f_ρ if there exist $\kappa > 0$ and $C_0 \geq 0$ such that*

$$Q_\rho^{(\rho)}(\tau, L) \geq \kappa P_4(\rho; \tau, L) - C_0 L$$

for sufficiently large τ and all $L \geq 1$.

Definition 12.2 (Localized no-hiding leakage)**.** *The localized leakage is subcritical if there exist $C_1 \geq 0$ and $\eta(\tau) \to 0$ such that*

$$Q_{\text{cross}}^{(\rho)}(\tau, L) + Q_{\text{rest}}^{(\rho)}(\tau, L) \geq -\eta(\tau) P_4(\rho; \tau, L) - C_1 L$$

for sufficiently large τ and all $L \geq 1$.

Proposition 12.3 (Localized bridge inequality)**.** *If localized witness coercivity and localized no-hiding leakage hold, then*

$$Q_{\text{bank}}^{(\rho)}(f; \tau, L) \geq (\kappa - \eta(\tau)) P_4(\rho; \tau, L) - (C_0 + C_1) L.$$

In particular, for sufficiently large τ,

$$Q_{\text{bank}}^{(\rho)}(f; \tau, L) \geq \frac{\kappa}{2} P_4(\rho; \tau, L) - (C_0 + C_1) L.$$

Proof. By the decomposition,

$$Q_{\text{bank}}^{(\rho)} = Q_\rho^{(\rho)} + Q_{\text{cross}}^{(\rho)} + Q_{\text{rest}}^{(\rho)}.$$

Using coercivity and no-hiding leakage,

$$Q_{\text{bank}}^{(\rho)} \geq \kappa P_4 - C_0 L - \eta(\tau) P_4 - C_1 L.$$

Hence

$$Q_{\text{bank}}^{(\rho)} \geq (\kappa - \eta(\tau)) P_4 - (C_0 + C_1) L.$$

Since $\eta(\tau) \to 0$, for sufficiently large τ,

$$\eta(\tau) \leq \frac{\kappa}{2}.$$

The second inequality follows. □

13 Bank-Subcritical Remainder Criterion

The following criterion explains why a C^2-type remainder bound with exponent $m_0 < 2$ is sufficient for bank-subcriticality.

Lemma 13.1 (Second-difference bound)**.** *Let*

$$h \in C^2([\tau - a, \tau + L + a]).$$

Then for every $u \in [\tau, \tau + L]$,

$$|W_a h(u)| \leq a^2 \sup_{v \in [u-a, u+a]} |h''(v)|.$$

Proof. By Taylor's theorem with integral remainder,

$$h(u + a) = h(u) + a h'(u) + \int_0^a (a - s) h''(u + s) \, \mathrm{d}s,$$

and

$$h(u - a) = h(u) - a h'(u) + \int_0^a (a - s) h''(u - s) \, \mathrm{d}s.$$

Adding and subtracting $2h(u)$, the linear terms cancel:

$$W_a h(u) = \int_0^a (a-s)\big(h''(u+s) + h''(u-s)\big)\ \mathrm{d}s.$$

Therefore

$$|W_a h(u)| \le 2\int_0^a (a-s)\ \mathrm{d}s \sup_{v\in[u-a,u+a]} |h''(v)| = a^2 \sup_{v\in[u-a,u+a]} |h''(v)|.$$

□

Proposition 13.2 (Integral bank-subcritical remainder criterion)**.** *Assume that for sufficiently large u,*

$$|f''_{\mathrm{rest}}(u)| \le C(1+u)^{m_0}$$

with

$$m_0 < 2.$$

Assume also that the localization window is bounded:

$$0 \le w_\rho(u) \le M_\rho.$$

Then, for fixed finite bank scales,

$$Q^{(\rho)}_{\mathrm{rest}}(\tau, L) \le C' \int_\tau^{\tau+L} (1+u)^{2m_0}\ \mathrm{d}u.$$

If the quartic obstruction satisfies the integral lower bound

$$P_4(\rho;\tau,L) \ge c\int_\tau^{\tau+L} u^4\ \mathrm{d}u - C_0 \int_\tau^{\tau+L} u^3\ \mathrm{d}u,$$

then, as the transported window moves to large scale,

$$\frac{Q^{(\rho)}_{\mathrm{rest}}(\tau,L)}{P_4(\rho;\tau,L)} \to 0.$$

In particular,

$$Q^{(\rho)}_{\mathrm{rest}} = o(P_4).$$

Proof. By the second-difference bound, for each fixed scale a_j,

$$|W_{a_j} f_{\mathrm{rest}}(u)| \le C_j(1+u)^{m_0}.$$

Therefore

$$|W_{a_j} f_{\mathrm{rest}}(u)|^2 \le C'_j(1+u)^{2m_0}.$$

Multiplying by the bounded localization window, summing over finitely many j, and integrating,

$$Q^{(\rho)}_{\mathrm{rest}}(\tau,L) \le C' \int_\tau^{\tau+L} (1+u)^{2m_0}\ \mathrm{d}u.$$

Since $2m_0 < 4$, the numerator has strictly lower polynomial order than

$$\int_\tau^{\tau+L} u^4\ \mathrm{d}u.$$

The correction

$$\int_\tau^{\tau+L} u^3\ \mathrm{d}u$$

is also lower order relative to the u^4-term at large scale. Hence

$$\frac{Q^{(\rho)}_{\mathrm{rest}}(\tau,L)}{P_4(\rho;\tau,L)} \to 0.$$

□

14 Uniform Gaussian Gram Coercivity: Status

Let a local cluster of ordinates be

$$\Gamma = \{\gamma_1, \dots, \gamma_N\}.$$

For $\alpha > 0$, define the Gaussian Gram matrix

$$K_{jk} = \exp\bigl(-\alpha(\gamma_j - \gamma_k)^2\bigr).$$

Lemma 14.1 (Finite-cluster Gaussian Gram positivity)**.** *If $\gamma_1, \dots, \gamma_N$ are distinct, then K is positive definite.*

Proof. The Gaussian kernel has the Fourier representation

$$e^{-\alpha(x-y)^2} = c_\alpha \int_{\mathbb{R}} e^{itx} e^{-ity} e^{-t^2/(4\alpha)}\, \mathrm{d}t$$

for a positive constant $c_\alpha > 0$. Thus, for any $c = (c_1, \dots, c_N) \in \mathbb{C}^N$,

$$\sum_{j,k=1}^{N} c_j \overline{c_k} K_{jk} = c_\alpha \int_{\mathbb{R}} \left| \sum_{j=1}^{N} c_j e^{it\gamma_j} \right|^2 e^{-t^2/(4\alpha)}\, \mathrm{d}t.$$

If this integral is zero, then

$$\sum_{j=1}^{N} c_j e^{it\gamma_j} = 0$$

for all real t. Since the γ_j are distinct, the exponentials are linearly independent. Therefore all $c_j = 0$. Hence K is positive definite. □

Open Problem 14.2 (Height-uniform local Gaussian Gram coercivity)**.** *After unfolding local zeta-zero clusters to unit mean spacing, prove a height-uniform lower bound*

$$\lambda_{\min}(K) \geq c(\alpha, L) > 0$$

for all sufficiently high local clusters inside a fixed unfolded window of size L.

Remark 14.3. *Finite-cluster positivity is not enough for the localized bridge. The needed statement is a height-uniform lower eigenvalue bound. Diagonal dominance can fail for moderately dense clusters, because nearby ordinates make off-diagonal Gaussian entries large. Therefore the desired estimate likely requires global structure of the unfolded zero configuration, not only pairwise separation.*

15 Master Conditional Contradiction

Theorem 15.1 (Fixed-channel contradiction)**.** *Fix a candidate zero*

$$\rho = \beta + i\gamma, \qquad \beta \neq \frac{1}{2}.$$

Assume A1–A4 hold for the same localized channel associated with ρ. Then a contradiction follows. Therefore no off-critical zero can satisfy these four statements simultaneously.

Proof. Since

$$\beta \neq \frac{1}{2},$$

we have

$$a_\rho = \frac{\beta - \frac{1}{2}}{2\pi} \neq 0.$$

By A1 and the quartic lower bound, for sufficiently large τ,

$$P_4(\rho;\tau,L) \geq c_\rho \tau^4 L.$$

By A3 and A4,

$$Q^{(\rho)}_{\text{bank}}(f;\tau,L) \geq (\kappa - \eta(\tau))P_4(\rho;\tau,L) - (C_0 + C_1)L.$$

For large τ, $\eta(\tau) \leq \kappa/2$. Hence

$$Q^{(\rho)}_{\text{bank}}(f;\tau,L) \geq \frac{\kappa}{2}P_4(\rho;\tau,L) - (C_0 + C_1)L.$$

Using the quartic lower bound,

$$Q^{(\rho)}_{\text{bank}}(f;\tau,L) \geq c'\tau^4 L - C''L.$$

After subtracting the baseline $E^{(\rho)}_0 L$ and dividing by L, the right-hand side is positive for sufficiently large τ. Therefore

$$E^{(\rho)}_{\text{ex}}(f) > 0.$$

But A2, by localized closure, gives

$$E^{(\rho)}_{\text{ex}}(f) = 0.$$

This is a contradiction. □

Corollary 15.2 (Conditional RH)**.** *If A1–A4 hold for every off-critical candidate zero, then every nontrivial zero of $\zeta(s)$ lies on the critical line.*

Proof. If an off-critical zero existed, the fixed-channel contradiction theorem would apply to its localized channel. This is impossible. Therefore no off-critical zero exists. □

16 What a Complete Proof Would Require

A complete unconditional proof would require exactly the following seven tasks on the genuine zeta residue channel.

1. **Transported drift law.** Derive

$$\delta_\rho(u) = a_\rho u + O(1), \qquad a_\rho = \frac{\beta - \frac{1}{2}}{2\pi},$$

   from a concrete transport normalization of the explicit formula.

2. **Genuine localized witness coercivity.** Upgrade the windowed-packet coercivity theorem to the actual transported zero-channel.

3. **Localized no-hiding leakage.** Prove

$$Q^{(\rho)}_{\text{cross}} + Q^{(\rho)}_{\text{rest}} \geq -\eta(\tau)P_4 - C_1 L, \qquad \eta(\tau) \to 0.$$

4. **Bank-subcritical remainder.** Establish residue-side regularity strong enough to imply

$$Q^{(\rho)}_{\text{rest}} = o(P_4).$$

5. **Uniform Gaussian Gram coercivity.** Prove height-uniform lower bounds for local Gaussian Gram matrices associated with actual zeta-zero clusters.

6. **Localized exact window law.** Verify the localized exact window law for the concrete dipole-regularized channel arising from the explicit formula.

7. **Model-to-zeta transfer.** Transfer the model estimates in the appendices to the genuine zeta residue channel.

These seven tasks are explicit. They are not claimed to be solved here.

17 Reality Check

17.1 Proved inside this document

The following statements are proved here:

- localized exact window law implies $E_{\mathrm{ex}}^{(\rho)}(f) = 0$;
- transported drift implies the quartic lower bound $P_4(\rho; \tau, L) \gtrsim \tau^4 L$;
- localized witness coercivity plus no-hiding leakage implies the bridge lower bound;
- a C^2 remainder with exponent $m_0 < 2$ is bank-subcritical;
- finite Gaussian Gram matrices of distinct ordinates are positive definite;
- pure exponential modes satisfy localized coercivity in a controlled model;
- windowed pure-exponential packets satisfy localized coercivity in a controlled model;
- coercivity is stable under small bank-energy perturbations;
- cross terms are quartic-subcritical under nonstationary phase assumptions;
- Gaussian heat-semigroup regularization sends tempered singular readouts to smooth profiles;
- the Source–Love Channel is a strict contraction meta-model with a unique fixed point.

17.2 Still open

The seven zeta-side tasks listed in the previous section remain open.

Remark 17.1. *The document is designed to be falsifiable at the bottlenecks. Failure of drift, coercivity, no-hiding, Gram control, or exact window closure on the genuine channel would block the RH implication.*

18 Outlook and Companion Status

The symbolic Source–Love geometry is useful for exposition:

$$\text{Source} = \text{drift-generating deviation}, \qquad \text{Love} = \text{closure-enforcing contraction},$$

and

$$\text{critical line} = \text{compatibility interface}.$$

This symbolic layer is not part of the proof chain. The detailed symbolic geometry belongs in a companion note, not in the main RH checkpoint.

The formal proof chain in this document remains exactly:

$$\text{transported drift}+\text{witness coercivity}+\text{no-hiding leakage}+\text{exact window} \Longrightarrow \text{fixed-channel contradiction}.$$

19 Conclusion

The localized transport–spectral program reduces the Riemann Hypothesis to a clean conditional contradiction.

The closure side forces

$$E_{\mathrm{ex}}^{(\rho)}(f) = 0.$$

The obstruction side, under transported drift and no-hiding, forces

$$E_{\mathrm{ex}}^{(\rho)}(f) > 0.$$

The contradiction is valid once the localized bridge package is verified on the genuine zeta residue channel.

This v3.2 checkpoint isolates the problem into explicit analytic bottlenecks and supplies rigorous model bricks showing that the required mechanisms are mathematically nonempty. The remaining work is hard, but it is now localized, modular, and precise.

Closure kills leakage.
Obstruction creates drift.
No-Hiding is the lock.
Localization makes the key visible.

Architecture is not proof.
But a sharp architecture tells us exactly what the proof must still accomplish.

A Soft Survival of a Transported Readout

This appendix records a structural compatibility lemma. Soft regularization does not create a transported drift, but it can prevent a pre-existing transported signal from being annihilated by a local hard readout collapse.

Lemma A.1 (Soft phase survival above baseline)**.** *Let $U \subset \mathbb{R}$ be a neighborhood of a local singular point. Let $\varepsilon > 0$ and define*

$$r_\varepsilon(\xi) = \sqrt{\xi^2 + \varepsilon^2}.$$

Assume that the effective readout admits an expansion

$$\Pi_{\eta,\varepsilon}(\xi) = \Pi_0 + \eta W(r_\varepsilon(\xi)) + R_{\eta,\varepsilon}(\xi), \qquad \xi \in U,$$

where:

(i) $\Pi_0 \in \mathbb{R}$*;*

(ii) $W : [0,\infty) \to [0,\infty)$ *is continuous and nonnegative;*

(iii) there exists $C > 0$ such that

$$|R_{\eta,\varepsilon}(\xi)| \le C\eta^2$$

for all $\xi \in U$ and sufficiently small $\eta > 0$;

(iv) there exists $w_* > 0$ such that

$$W(r_\varepsilon(\xi)) \ge w_*$$

for all $\xi \in U$.

Then there exists $\eta_0 > 0$ such that for all $0 < \eta < \eta_0$,

$$\Pi_{\eta,\varepsilon}(\xi) \geq \Pi_0 + \frac{1}{2}\eta w_* > \Pi_0$$

for all $\xi \in U$. If $\Pi_0 = 0$, then

$$\Pi_{\eta,\varepsilon}(\xi) > 0.$$

Proof. Using the expansion,

$$\Pi_{\eta,\varepsilon}(\xi) = \Pi_0 + \eta W(r_\varepsilon(\xi)) + R_{\eta,\varepsilon}(\xi).$$

By the lower bound on W and the remainder estimate,

$$\Pi_{\eta,\varepsilon}(\xi) \geq \Pi_0 + \eta w_* - C\eta^2.$$

Choose $\eta_0 > 0$ such that

$$C\eta^2 \leq \frac{1}{2}\eta w_*$$

for $0 < \eta < \eta_0$. Then

$$\Pi_{\eta,\varepsilon}(\xi) \geq \Pi_0 + \frac{1}{2}\eta w_*.$$

If $\Pi_0 = 0$, this is strictly positive. □

Corollary A.2 (Compatibility with transported drift)**.** *Let f_ρ be a localized transported mode associated with an off-critical zero*

$$\rho = \beta + i\gamma, \qquad \beta \neq \frac{1}{2}.$$

Suppose that in a transported frame its effective phase variable satisfies

$$\delta_\rho(u) = a_\rho u + r_\rho(u), \qquad a_\rho = \frac{\beta - \frac{1}{2}}{2\pi},$$

with r_ρ bounded. If the local readout near a hard singularity satisfies the soft expansion in the preceding lemma, then local loss of hard sign-readout does not force annihilation of the transported signal in the soft channel.

Proof. Apply the previous lemma to the local readout of the transported channel. □

Remark A.3. *This result does not prove the transported drift law for the genuine zeta function. It proves only that if a transported drift is already present, soft regularization can keep it visible through a local hard readout collapse.*

B Pure Exponential Coercivity Model

This appendix proves localized witness coercivity in the exact pure-exponential model.

Proposition B.1 (Localized witness coercivity: pure exponential model)**.** *Let*

$$\rho = \beta + i\gamma, \qquad 0 < \beta < 1, \qquad \beta \neq \frac{1}{2}.$$

Define

$$g_\rho(u) = A_\rho e^{\rho u}, \qquad A_\rho = \exp\left(\frac{\rho^2}{4\pi}\right).$$

Assume the transported drift law. Let $w_{\rho,\tau,L}$ be a plateau window satisfying

$$0 < m \leq w_{\rho,\tau,L}(u) \leq M < \infty$$

for all $u \in [\tau, \tau + L]$. Then there exist constants

$$\kappa > 0, \qquad C_0 \geq 0, \qquad \tau_0 > 0$$

such that for all $\tau \geq \tau_0$ and $L \geq 1$,

$$Q^{(\rho)}_{\text{bank}}(g_\rho; \tau, L) \geq \kappa P_4(\rho; \tau, L) - C_0 L.$$

Proof. Since g_ρ is a pure exponential,

$$W_{a_j} g_\rho(u) = g_\rho(u + a_j) - 2g_\rho(u) + g_\rho(u - a_j) = d_{\rho,a_j} g_\rho(u),$$

where

$$d_{\rho,a_j} = e^{\rho a_j} - 2 + e^{-\rho a_j}.$$

Hence

$$Q^{(\rho)}_{\text{bank}}(g_\rho; \tau, L) = |A_\rho|^2 \sum_{j=0}^{J} w_j^2 |d_{\rho,a_j}|^2 \int_\tau^{\tau+L} w_{\rho,\tau,L}(u) e^{2\beta u} \, \mathrm{d}u.$$

Set

$$S_\rho = \sum_{j=0}^{J} w_j^2 |d_{\rho,a_j}|^2.$$

We claim $S_\rho > 0$. If $d_{\rho,a_j} = 0$, then

$$e^{\rho a_j} + e^{-\rho a_j} = 2,$$

so

$$\cosh(\rho a_j) = 1.$$

The solutions of $\cosh z = 1$ are $z = 2\pi i k$, $k \in \mathbb{Z}$, hence have zero real part. But

$$\Re(\rho a_j) = \beta a_j > 0.$$

Thus $d_{\rho,a_j} \neq 0$, and therefore $S_\rho > 0$.

Using $w_{\rho,\tau,L} \geq m$,

$$Q^{(\rho)}_{\text{bank}}(g_\rho; \tau, L) \geq |A_\rho|^2 S_\rho m \int_\tau^{\tau+L} e^{2\beta u} \, \mathrm{d}u.$$

Therefore

$$Q^{(\rho)}_{\text{bank}}(g_\rho; \tau, L) \geq c_1 \left(e^{2\beta(\tau+L)} - e^{2\beta\tau} \right),$$

where

$$c_1 = |A_\rho|^2 S_\rho \frac{m}{2\beta} > 0.$$

By the polynomial upper bound for P_4,

$$P_4(\rho; \tau, L) \leq C_\rho \left((\tau + L)^5 - \tau^5 + L \right).$$

The exponential lower bound dominates the polynomial upper bound for sufficiently large τ, uniformly for $L \geq 1$. Hence, choosing $\kappa > 0$ small and τ_0 large,

$$Q^{(\rho)}_{\text{bank}}(g_\rho; \tau, L) \geq \kappa P_4(\rho; \tau, L) - C_0 L.$$

□

Remark B.2. *This model is intentionally idealized. It does not prove coercivity for the genuine zeta residue channel.*

C Windowed Packet Coercivity and Perturbative Stability

This appendix is the main technical upgrade beyond the pure exponential model. It shows that coercivity survives localization by a packet window with an interior plateau.

Let

$$A_* = \max_{0 \le j \le J} a_j.$$

Definition C.1 (Admissible packet window)**.** *An admissible packet window $\chi_{\tau,L} \in C^2(\mathbb{R})$ satisfies:*

$$0 \le \chi_{\tau,L} \le 1,$$

$$\chi_{\tau,L}(u) = 1 \quad \text{for } u \in [\tau + A_*,\, \tau + L - A_*],$$

and

$$\operatorname{supp} \chi_{\tau,L} \subset [\tau, \tau + L].$$

We assume $L \ge 2A_* + 1$, so the interior plateau has length at least 1.

Definition C.2 (Windowed pure-exponential packet)**.** *Define*

$$h_{\rho,\tau,L}(u) = \chi_{\tau,L}(u) A_\rho e^{\rho u}.$$

Lemma C.3 (Interior exactness of the second-difference witness)**.** *If*

$$u \in [\tau + A_*,\, \tau + L - A_*],$$

then for every bank scale a_j,

$$W_{a_j} h_{\rho,\tau,L}(u) = d_{\rho,a_j} A_\rho e^{\rho u},$$

where

$$d_{\rho,a_j} = e^{\rho a_j} - 2 + e^{-\rho a_j}.$$

Proof. If

$$u \in [\tau + A_*,\, \tau + L - A_*],$$

then

$$u - a_j,\ u,\ u + a_j \in [\tau, \tau + L]$$

and, more precisely, all three points lie inside the plateau region where

$$\chi_{\tau,L} = 1.$$

Therefore

$$h_{\rho,\tau,L}(u \pm a_j) = A_\rho e^{\rho(u \pm a_j)}$$

and

$$h_{\rho,\tau,L}(u) = A_\rho e^{\rho u}.$$

Hence

$$W_{a_j} h_{\rho,\tau,L}(u) = A_\rho e^{\rho u}(e^{\rho a_j} - 2 + e^{-\rho a_j}) = d_{\rho,a_j} A_\rho e^{\rho u}.$$

□

Theorem C.4 (Localized packet coercivity)**.** *Let*

$$\rho = \beta + i\gamma, \qquad 0 < \beta < 1, \qquad \beta \ne \frac{1}{2}.$$

Assume the transported drift law. Let $h_{\rho,\tau,L}$ be the windowed packet above, with $L \ge 2A_* + 1$. Then there exist constants

$$\kappa > 0, \qquad C_0 \ge 0, \qquad \tau_0 > 0$$

such that for all $\tau \ge \tau_0$,

$$Q^{(\rho)}_{\text{bank}}(h_{\rho,\tau,L}; \tau, L) \ge \kappa P_4(\rho; \tau, L) - C_0 L.$$

Proof. Because the integrand in the bank energy is nonnegative, we may restrict the integral to the interior plateau:

$$Q^{(\rho)}_{\text{bank}}(h_{\rho,\tau,L};\tau,L) \geq \sum_{j=0}^{J} w_j^2 \int_{\tau+A_*}^{\tau+L-A_*} w_\rho(u)|W_{a_j} h_{\rho,\tau,L}(u)|^2 \, \mathrm{d}u.$$

Assume the localization weight is bounded below on the relevant packet region:

$$w_\rho(u) \geq m_\rho > 0$$

on the interior plateau. Using interior exactness,

$$|W_{a_j} h_{\rho,\tau,L}(u)|^2 = |d_{\rho,a_j}|^2 |A_\rho|^2 e^{2\beta u}.$$

Therefore

$$Q^{(\rho)}_{\text{bank}}(h_{\rho,\tau,L};\tau,L) \geq m_\rho |A_\rho|^2 \left(\sum_{j=0}^{J} w_j^2 |d_{\rho,a_j}|^2 \right) \int_{\tau+A_*}^{\tau+L-A_*} e^{2\beta u} \, \mathrm{d}u.$$

As in the pure exponential model,

$$S_\rho = \sum_{j=0}^{J} w_j^2 |d_{\rho,a_j}|^2 > 0.$$

Hence

$$Q^{(\rho)}_{\text{bank}}(h_{\rho,\tau,L};\tau,L) \geq C_\rho \int_{\tau+A_*}^{\tau+L-A_*} e^{2\beta u} \, \mathrm{d}u$$

for some $C_\rho > 0$.

The plateau has length at least 1, so the exponential integral is positive. More explicitly,

$$\int_{\tau+A_*}^{\tau+L-A_*} e^{2\beta u} \, \mathrm{d}u = \frac{e^{2\beta(\tau+L-A_*)} - e^{2\beta(\tau+A_*)}}{2\beta}.$$

This exponential quantity dominates the polynomial upper bound

$$P_4(\rho;\tau,L) \leq C \left((\tau+L)^5 - \tau^5 + L \right)$$

for sufficiently large τ, uniformly for $L \geq 2A_* + 1$. Therefore, choosing $\kappa > 0$ small and τ_0 large enough,

$$Q^{(\rho)}_{\text{bank}}(h_{\rho,\tau,L};\tau,L) \geq \kappa P_4(\rho;\tau,L) - C_0 L.$$

□

Remark C.5. *The proof does not need to estimate boundary layers from below. Since the bank energy is nonnegative, the boundary contribution can be ignored. The coercive contribution comes entirely from the interior plateau where the second-difference witness is exact.*

Lemma C.6 (Quadratic lower stability)**.** *For all $x, y \in \mathbb{C}$,*

$$|x+y|^2 \geq \frac{1}{2}|x|^2 - |y|^2.$$

Proof. Using

$$|x| = |(x+y) - y| \leq |x+y| + |y|,$$

we have

$$|x|^2 \leq 2|x+y|^2 + 2|y|^2.$$

Rearranging gives

$$|x+y|^2 \geq \frac{1}{2}|x|^2 - |y|^2.$$

□

Proposition C.7 (Perturbative stability of coercivity)**.** *Suppose a model mode g satisfies*

$$Q^{(\rho)}_{\text{bank}}(g;\tau,L) \geq \kappa P_4(\rho;\tau,L) - C_0 L.$$

Let e be a perturbation satisfying

$$Q^{(\rho)}_{\text{bank}}(e;\tau,L) \leq \frac{\kappa}{4} P_4(\rho;\tau,L) + C_e L.$$

Then

$$Q^{(\rho)}_{\text{bank}}(g+e;\tau,L) \geq \frac{\kappa}{4} P_4(\rho;\tau,L) - \left(\frac{C_0}{2} + C_e\right) L.$$

Proof. Apply the quadratic lower stability inequality pointwise with

$$x = W_{a_j} g(u), \qquad y = W_{a_j} e(u).$$

After multiplying by $w_j^2 w_\rho(u)$, summing over j, and integrating,

$$Q^{(\rho)}_{\text{bank}}(g+e;\tau,L) \geq \frac{1}{2} Q^{(\rho)}_{\text{bank}}(g;\tau,L) - Q^{(\rho)}_{\text{bank}}(e;\tau,L).$$

Using the two hypotheses,

$$Q^{(\rho)}_{\text{bank}}(g+e;\tau,L) \geq \frac{1}{2}(\kappa P_4 - C_0 L) - \left(\frac{\kappa}{4} P_4 + C_e L\right).$$

Thus

$$Q^{(\rho)}_{\text{bank}}(g+e;\tau,L) \geq \frac{\kappa}{4} P_4 - \left(\frac{C_0}{2} + C_e\right) L.$$

□

Remark C.8. *This identifies the next genuine zeta-side target: prove that the difference between the actual transported zero-channel and a controlled packet model is small in bank energy relative to P_4.*

D Nonstationary Cross-Term Decay

This appendix proves a model nonstationary-phase estimate for cross-term decay.

Lemma D.1 (Localized cross-term decay by nonstationary phase)**.** *Define*

$$Q^{(\rho)}_{\text{cross}}(\tau,L) = 2\sum_{j=0}^{J} w_j^2 \operatorname{Re} \int_\tau^{\tau+L} w_\rho(u) B_{\rho,j}(u) \overline{B_{\text{rest},j}(u)} e^{i\Phi_j(u)}\, \mathrm{d}u.$$

Assume that, on $[\tau, \tau+L]$,

$$\|w_\rho\|_\infty + \|w'_\rho\|_\infty \leq C_w,$$
$$|B_{\rho,j}(u)| + |B'_{\rho,j}(u)| \leq C_\rho \tau^2,$$
$$|B_{\text{rest},j}(u)| + |B'_{\text{rest},j}(u)| \leq C_{\text{rest}},$$
$$|\Phi'_j(u)| \geq c_\Phi \tau, \qquad |\Phi''_j(u)| \leq C_\Phi.$$

Then

$$|Q^{(\rho)}_{\text{cross}}(\tau,L)| \leq C_{\text{cross}} \tau L$$

for sufficiently large τ and all $L \geq 1$.

Proof. Set

$$g_j(u) = w_\rho(u) B_{\rho,j}(u) \overline{B_{\mathrm{rest},j}(u)}.$$

By the product rule and the assumed bounds,

$$|g_j(u)| + |g_j'(u)| \le C_g \tau^2.$$

Let

$$I_j = \int_\tau^{\tau+L} g_j(u) e^{i\Phi_j(u)} \, \mathrm{d}u.$$

Since Φ_j' is bounded away from zero,

$$e^{i\Phi_j(u)} = \frac{1}{i\Phi_j'(u)} \frac{\mathrm{d}}{\mathrm{d}u} e^{i\Phi_j(u)}.$$

Integration by parts gives

$$I_j = \left[\frac{g_j(u)}{i\Phi_j'(u)} e^{i\Phi_j(u)} \right]_\tau^{\tau+L} - \int_\tau^{\tau+L} \left(\frac{g_j'(u)}{i\Phi_j'(u)} - \frac{g_j(u)\Phi_j''(u)}{i(\Phi_j'(u))^2} \right) e^{i\Phi_j(u)} \, \mathrm{d}u.$$

Taking absolute values,

$$|I_j| \le 2 \sup \frac{|g_j|}{|\Phi_j'|} + \int_\tau^{\tau+L} \frac{|g_j'|}{|\Phi_j'|} \, \mathrm{d}u + \int_\tau^{\tau+L} \frac{|g_j||\Phi_j''|}{|\Phi_j'|^2} \, \mathrm{d}u.$$

Using the hypotheses,

$$\frac{|g_j|}{|\Phi_j'|} \le C\tau,$$

$$\frac{|g_j'|}{|\Phi_j'|} \le C\tau,$$

and

$$\frac{|g_j||\Phi_j''|}{|\Phi_j'|^2} \le C.$$

Thus

$$|I_j| \le C\tau + C\tau L + CL \le C'\tau L$$

for $\tau \ge 1$, $L \ge 1$. Summing over finitely many j gives the result. □

Corollary D.2 (Cross term is quartic-subcritical)**.** *If*

$$P_4(\rho; \tau, L) \ge c_4 \tau^4 L,$$

then

$$\frac{|Q_{\mathrm{cross}}^{(\rho)}(\tau, L)|}{P_4(\rho; \tau, L)} \le \frac{C_{\mathrm{cross}}}{c_4 \tau^3} \to 0.$$

Thus

$$Q_{\mathrm{cross}}^{(\rho)} = o(P_4).$$

Proof. Combine

$$|Q_{\mathrm{cross}}^{(\rho)}(\tau, L)| \le C_{\mathrm{cross}} \tau L$$

with

$$P_4(\rho; \tau, L) \ge c_4 \tau^4 L.$$

□

E Gaussian Hammer: Heat-Semigroup Regularization

This appendix records the Gaussian Hammer principle: Gaussian readout converts singular or distributional data into a smooth effective profile and suppresses high frequencies exponentially.

We use the Fourier convention

$$\widehat{f}(\xi) = \int_{\mathbb{R}^n} f(x) e^{-2\pi i x\cdot\xi}\, \mathrm{d}x.$$

For $\sigma > 0$, define

$$G_\sigma(x) = \sigma^{-n} e^{-\pi|x|^2/\sigma^2}.$$

Lemma E.1 (Fourier transform of the scaled Gaussian)**.** *With the above convention,*

$$\widehat{G_\sigma}(\xi) = e^{-\pi\sigma^2|\xi|^2}.$$

Proof. The standard identity is

$$\widehat{e^{-\pi|x|^2}}(\xi) = e^{-\pi|\xi|^2}.$$

Since

$$G_\sigma(x) = \sigma^{-n} G_1(x/\sigma),$$

the Fourier scaling rule gives

$$\widehat{G_\sigma}(\xi) = \widehat{G_1}(\sigma\xi) = e^{-\pi\sigma^2|\xi|^2}.$$

□

Theorem E.2 (Gaussian hammer regularizes singular readout)**.** *Let $u \in \mathcal{S}'(\mathbb{R}^n)$ be a tempered distribution and define*

$$u_\sigma = G_\sigma * u.$$

Then:

(1) $u_\sigma \in C^\infty(\mathbb{R}^n)$*;*

(2)

$$\widehat{u_\sigma}(\xi) = e^{-\pi\sigma^2|\xi|^2}\widehat{u}(\xi);$$

(3) high-frequency components are exponentially suppressed by

$$e^{-\pi\sigma^2|\xi|^2};$$

(4) if $u = \delta_{x_0}$, then

$$u_\sigma(x) = G_\sigma(x - x_0).$$

Proof. Since $G_\sigma \in \mathcal{S}(\mathbb{R}^n)$ and $u \in \mathcal{S}'(\mathbb{R}^n)$, convolution with G_σ produces a smooth function. For every multiindex α,

$$\partial^\alpha u_\sigma = \partial^\alpha(G_\sigma * u) = (\partial^\alpha G_\sigma) * u,$$

which is well-defined and continuous.

The Fourier convolution identity gives

$$\widehat{u_\sigma} = \widehat{G_\sigma}\widehat{u} = e^{-\pi\sigma^2|\xi|^2}\widehat{u}.$$

The exponential multiplier suppresses high frequencies.

Finally,

$$G_\sigma * \delta_{x_0} = G_\sigma(\cdot - x_0),$$

so

$$u_\sigma(x) = G_\sigma(x - x_0).$$

□

Corollary E.3 (Gaussian suppression beats every polynomial)**.** *For every $N > 0$, there exists $C_{N,\sigma} > 0$ such that*

$$e^{-\pi\sigma^2|\xi|^2} \le C_{N,\sigma}(1+|\xi|)^{-N}.$$

Proof. Let $r = |\xi|$. The function

$$r \mapsto e^{-\pi\sigma^2 r^2}(1+r)^N$$

is continuous on $[0,\infty)$ and tends to 0 as $r \to \infty$. Hence it is bounded. □

Proposition E.4 (Gaussian readout as heat flow)**.** *Let Δ be the Laplacian on $\mathbb{R}^n$. Then*

$$G_\sigma * u = e^{t\Delta}u, \qquad t = \frac{\sigma^2}{4\pi}.$$

Proof. The Fourier transform of the Laplacian is

$$\widehat{\Delta u}(\xi) = -4\pi^2|\xi|^2\widehat{u}(\xi).$$

Thus

$$\widehat{e^{t\Delta}u}(\xi) = e^{-4\pi^2 t|\xi|^2}\widehat{u}(\xi).$$

Choosing

$$t = \frac{\sigma^2}{4\pi}$$

gives

$$e^{-4\pi^2 t|\xi|^2} = e^{-\pi\sigma^2|\xi|^2},$$

which is exactly the Fourier multiplier for convolution by G_σ. □

Remark E.5. *This does not prove removal of physical singularities in general relativity. It proves the analytic readout statement:*

$$\textit{singular raw source} \longmapsto \textit{smooth Gaussian effective profile.}$$

F Source–Love Channel: Companion Meta-Model

This appendix is not used as an input to the RH contradiction. It records a separate operator model for the general principle that stable structure can arise from the interaction of transport, controlled perturbation, and contraction.

Let

$$\mathcal{H} \simeq \mathbb{C}^d,$$

and let $\mathcal{D}(\mathcal{H})$ denote the set of density operators on $\mathcal{H}$. Fix:

- a unitary operator $U : \mathcal{H} \to \mathcal{H}$;
- a target state $\rho_\infty \in \mathcal{D}(\mathcal{H})$;
- a source state $\rho_F \in \mathcal{D}(\mathcal{H})$;
- parameters
$$\lambda \in (0,1], \qquad \gamma \in [0,1).$$

Define the Source–Love Channel by

$$\mathcal{T}(\rho) = (1-\lambda)(1-\gamma)U\rho U^\dagger + (1-\lambda)\gamma U\rho_F U^\dagger + \lambda\rho_\infty.$$

The three terms encode:

$$\text{transport} + \text{source perturbation} + \text{stabilizing contraction}.$$

Theorem F.1 (Source–Love contraction)**.** *For all $\rho, \sigma \in \mathcal{D}(\mathcal{H})$,*

$$\|\mathcal{T}(\rho) - \mathcal{T}(\sigma)\|_1 = (1-\lambda)(1-\gamma)\|\rho - \sigma\|_1.$$

Hence $\mathcal{T}$ is a strict contraction whenever

$$q = (1-\lambda)(1-\gamma) < 1.$$

Proof. Subtract:

$$\mathcal{T}(\rho) - \mathcal{T}(\sigma) = (1-\lambda)(1-\gamma)U(\rho - \sigma)U^\dagger.$$

The trace norm is unitarily invariant:

$$\|U(\rho - \sigma)U^\dagger\|_1 = \|\rho - \sigma\|_1.$$

Therefore

$$\|\mathcal{T}(\rho) - \mathcal{T}(\sigma)\|_1 = (1-\lambda)(1-\gamma)\|\rho - \sigma\|_1.$$

□

Corollary F.2 (Unique stable attractor)**.** *The map $\mathcal{T}$ has a unique fixed point $\rho_* \in \mathcal{D}(\mathcal{H})$, and*

$$\|\mathcal{T}^n(\rho) - \rho_*\|_1 \le q^n \|\rho - \rho_*\|_1.$$

Proof. This is the Banach fixed-point theorem applied to the complete metric space

$$(\mathcal{D}(\mathcal{H}), \|\cdot\|_1).$$

□

F.1 Fixed point via Neumann series

Let

$$c = (1-\lambda)(1-\gamma),$$

and define

$$b = (1-\lambda)\gamma U \rho_F U^\dagger + \lambda \rho_\infty.$$

Then the fixed point equation is

$$\rho_* = cU\rho_* U^\dagger + b.$$

Equivalently,

$$(I - c\,\mathrm{Ad}_U)\rho_* = b,$$

where

$$\mathrm{Ad}_U(X) = UXU^\dagger.$$

Proposition F.3 (Neumann-series representation)**.** *The unique fixed point satisfies*

$$\rho_* = \sum_{n=0}^{\infty} c^n \,\mathrm{Ad}_U^n(b).$$

Proof. Since

$$\|\,\mathrm{Ad}_U\,\|_{1\to 1} = 1$$

and $0 \le c < 1$, the operator $I - c\,\mathrm{Ad}_U$ is invertible by the Neumann series:

$$(I - c\,\mathrm{Ad}_U)^{-1} = \sum_{n=0}^{\infty} c^n \,\mathrm{Ad}_U^n\,.$$

Applying this to b gives the result. □

F.2 Progress as reduction of epistemic debt

Define the epistemic debt

$$D(\rho) = \|\rho - \rho_*\|_1.$$

Then

$$D(\mathcal{T}\rho) = qD(\rho).$$

Definition F.4 (Progress functional)**.** *Define*

$$\mathcal{P}(\rho) = D(\rho) - D(\mathcal{T}\rho).$$

Hence

$$\mathcal{P}(\rho) = (1-q)D(\rho).$$

Theorem F.5 (Progress is positive away from the attractor)**.** *If $\rho \neq \rho_*$, then*

$$\mathcal{P}(\rho) > 0.$$

If $\rho = \rho_*$, then

$$\mathcal{P}(\rho) = 0.$$

Proof. Since $1 - q > 0$,

$$\mathcal{P}(\rho) = (1-q)\|\rho - \rho_*\|_1.$$

This is positive exactly when $\rho \neq \rho_*$. □

F.3 Continuous-time limit

Let

$$\lambda = \kappa\Delta t, \qquad \gamma = \mu\Delta t, \qquad U = e^{-iH\Delta t}.$$

Then

$$U\rho U^\dagger = \rho - i\Delta t[H,\rho] + o(\Delta t).$$

Proposition F.6 (Continuous-time Lindblad-type limit)**.** *In the limit $\Delta t \to 0$, the Source–Love dynamics converge formally to*

$$\frac{\mathrm{d}\rho}{\mathrm{d}t} = -i[H,\rho] + \kappa(\rho_\infty - \rho) + \mu(\rho_F - \rho).$$

Proof. Substitute

$$\lambda = \kappa\Delta t, \qquad \gamma = \mu\Delta t, \qquad U = e^{-iH\Delta t}$$

into $\mathcal{T}$. Keeping only first-order terms gives

$$\mathcal{T}(\rho) - \rho = \Delta t\,(-i[H,\rho] + \kappa(\rho_\infty - \rho) + \mu(\rho_F - \rho)) + o(\Delta t).$$

Dividing by Δt and taking $\Delta t \to 0$ gives the generator. □

Remark F.7. *The Source–Love Channel is not part of the RH proof chain. It is a companion model for the general principle that stable readout can arise from coherent transport, controlled injection, and contraction.*

Progress is not doing everything possible.

Progress is converting epistemic debt into stable readout.

Love is not sentiment; it is the contraction that lets exploration survive without exploding.